

Your analysis plan, in one hour

Participant worksheet. Work on YOUR project. Take it home finished.

What you leave this session with

A **written analysis plan for your own operational research project** — the tables and figures you will produce, and the denominator for each one. Written down, before you see your results. This is the document that protects you from every trap in Factsheets 2, 3 and 7.

1 State your research question, again, in one sentence 5 min

Not the topic — the *question*. If it does not contain a population, an outcome and a time frame, it is not finished yet.

2 List every variable you will actually use 10 min

Only the ones that answer the question above. If a variable does not appear in a planned table or figure, you do not need it — and collecting it is costing someone time.

Variable name	What it means	Type (nominal / ordinal / discrete / continuous)	How you will summarise it

3 Decide your denominators NOW 10 min

For each main result, write down exactly who is included and who is excluded — before you calculate anything. See Factsheet 3.

Result I will report	Numerator = who?	Denominator = who?	Who is excluded, and why

Result I will report	Numerator = who?	Denominator = who?	Who is excluded, and why

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Sketch your Table 1 10 min

Every operational research paper has a Table 1: the characteristics of your study population. Draw it by hand, with the row labels filled in and the numbers left blank. You are designing the shape of your result before you have it.

Table 1 — sketch

5

Sketch your main results table or figure 10 min

This is the one that answers your research question. Draw it empty. If you cannot draw it, your question is not yet answerable with the data you have — and it is far better to discover that today than in six months.

Main result — sketch

6

The missing data question 5 min

Every dataset has gaps. Write down now how you will handle them, so you are not tempted to decide later based on which answer you prefer.

- Which variables do you expect to have missing data?
- Will you exclude those records, or report 'unknown' as its own category?
- How will you report the amount of missing data in your paper?

7

Present to your mentor 10 min

Show your mentor the two sketches and your denominator table. Ask them one question: **“What have I missed?”**

Check before you leave the room

- ☐ My research question fits in one sentence and names a population, an outcome and a time frame.
- ☐ Every variable in my list has a type written next to it.
- ☐ Every planned result has a denominator written down.
- ☐ I have sketched Table 1 and my main result.
- ☐ I have decided how I will handle missing data.
- ☐ My mentor has seen it and I have written down what they said.